



Iarnród Éireann
Irish Rail

Power Platform - Portfolio Project Management Solution

Invitation to Tender

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1 Introduction

1.1 Company Background

Iarnród Éireann (IE) is the national railway operator in the Republic of Ireland which operates a mix of Intercity and Commuter services, servicing 146 stations and employing circa 5,000 staff nationwide.

Intercity services mainly operate from Dublin and link the main cities of Ireland including Cork, Limerick, Tralee, Waterford, Galway, Westport, Sligo, Belfast and Rosslare Europort. An extensive Commuter service operates within the Dublin region and surrounding counties, while Cork and Limerick also have commuter services on a smaller scale.

IE is also responsible for the maintenance and development of the rail infrastructure including the track, overhead lines, stations and the rail fleet required to operate the service. IE also operates Port services at Rosslare Europort.

With safety as our number one corporate goal, IE's corporate identify and corporate objectives is critically dependant on its people, with their skills, training, professionalism, adaptability, organisation and efficiency all considered central to our success.

IE is largely separated into Railway Undertaking (IERU) which manage the operations of the rolling stock, Infrastructure Management (IEIM) which operate and maintain and Capital Investments (CI) which design and build railway assets and associated infrastructure (Buildings and Stations).

CI is the division within IE which plans and delivers infrastructure and fleet investment on the IE rail network. CI undertakes the construction of large-scale multi-year infrastructure projects and can vary from small bridge and rail upgrades, up to multi-year programmes. These projects entail the design and tendering early phases of construction through to the building and commissioning aspects associated with safety critical infrastructure projects.

1.2 Invitation to Tender (ITT)

IE CI invites qualified and experienced service providers to submit tenders for the provision of a Project, Programme, and Portfolio Management (PPM) Solution built utilising IE's Microsoft Power Platform. The objective of the PPM solution is the following:

- Provide a centralised platform for project, programme, and portfolio oversight, providing a single source of the truth for portfolio performance.
- Standardise governance, assurance, and reporting across teams.
- Standardise data collection by introducing consistent data-entry forms, workflows, and structures across project and portfolio reporting.
- Improve visibility of delivery performance, risks, financials etc.
- Support data-driven decision making through dashboards and other reporting metrics.
- Enable scalable functionality aligned with future organisational needs.

The successful supplier will provide and configure predeveloped Power Platform templates and/or components to rapidly establish a digital PPM data capture and reporting solution. The core requirement is the quick delivery of a Minimum Viable Product (MVP) with the ability to scale the solution. The PPM MVP solution must provide an option to expand licensing, functionality, and integrations in future deployments.

The predeveloped Power Platform templates and/or components must support a future full PPM solution deployment with the following **core capabilities**:

- Centralised Project, Programme & Portfolio Management
- Governance, Assurance & Compliance
- Schedule & Programme Management
- Financial & Funding Management
- Risk & Issue Management
- Change Management
- Benefits Management
- Capacity & Resource Management
- Timesheet Management
- Reporting & Analytics
- Configurable Workflows
- User Access & Security

Additionally, the PPM solution will require integrations with existing IÉ ICT software / infrastructure. A breakdown of these integration requirements as provided within the detailed requirements of this document.

2 Background and Solution Requirements

2.1 About Iarnród Éireann Capital Investments

IE CI manages a large portfolio of projects ranging in size, scale and complexity. CI have experienced a significant increase in the volume of large capital projects, programmes and increased value of contracts. CI are taking appropriate steps to transition from a static, fragmented and manual ways of working to an integrated, centralised, scalable and intelligent best-practice digital environment.

CI's vision is to develop a best-in-class Project Management Office (PMO) Project Controls & Reporting function, ensuring accurate reporting and integrated project, programme and portfolio management. To create a digitally enabled ecosystem that delivers consistent project controls, improved process and system integration, real-time metrics, trends and insights, improved information management and robust governance and assurance. The objective of these initiatives is to provide the CI leadership with a single source of truth and confidence in the reporting and data being generated. The transition to a digital ecosystem will enable automated reporting, streamlined workflows, improved performance tracking, dynamic risk integration and schedule optimisation.

2.2 Users

CI have a requirement to categorise users of the solution into three broad classifications, System Administrator, Standard User and Team Member User.

- System Administrator – to manage all system settings, security roles, data structures, and overall PPM platform configuration.
- Standard - access and update programmes and/or projects relevant to their role, with permissions to view and interact with core PPM capabilities.
- Team Member – limited access to certain PPM capabilities but allows the user input for collaboration purposes. e.g. providing updates to project tasks or issues.

The expected number of users is listed below, with the expected increase over the life of the contract.

Note: the user numbers provided are indicative only and are intended to illustrate the anticipated scale of system usage. Actual user volumes may vary and will be adjusted in line with evolving business requirements

User Type	Pilot	2026	2029
System Administrator	2	5	5
Standard	18	100	150
Team Member	0	100	150

2.3 Scope of Work

This Scope of Works defines the phased delivery of a PPM solution built on Microsoft Power Platform using pre-developed templates and/or components. The delivery model is designed to provide flexibility, scalability, and full cost transparency. It shall allow CI to evaluate the solution before committing to full deployment and long-term support.

The project will be delivered in three phases:

- MVP Pilot Deployment
- Solution Enhancements (Optional)
- Full Deployment + Post Deployment Support (Optional)

System Enhancements, Full Deployment and Post Deployment Support elements of this tender will only be triggered upon successful completion and evaluation of the MVP Pilot.

The objectives of this approach are as follows:

- Deploy a scalable and user-friendly PPM system built on IÉ's own Microsoft Power Platform.
- Allow CI to validate suitability through a controlled MVP Pilot.
- Provide flexibility for CI to adopt the solution in its 'vanilla' configuration or proceed with enhancements based on pilot findings.
- Establish an optional long-term support framework.

MVP Pilot Deployment

The provision of an MVP using pre-built Power Platform PPM templates and/or components with minimal "vanilla" configuration. The objective is to validate core functionality and assess whether further development is required.

The MVP will be deployed to a pilot group of up to 20 users, supporting up to 10 live projects over approx. 3 reporting cycles (4 weekly periods).

As part of this pilot the vendor is required to provide the following:

- Licencing
 - Provision of licences for 20 users (templates and/or components licencing).
 - Licences shall cover the full functionality of the vanilla 'out of box' solution.
 - The solution shall be deployed into an IÉ managed Microsoft Power Platform environment. All platform hosting, infrastructure, and Microsoft licensing requirements are the responsibility of IÉ and are expressly excluded from the scope of this tender. This includes, but is not limited to, the provision of a Dev/Test and Production Power Platform environments, Microsoft Power Apps Premium licensing and Power BI Premium/Pro licensing. The supplier is required to deliver and configure the solution within the IÉ environment only and will not be responsible for providing, managing, or procuring any underlying platform licences or hosting services.

- MVP Pilot Deployment & Technical Services
 - Implementation Approach & Plan
 - Provide IÉ their solutions Power Platform requirements (Environment Specs, Licences etc.)
 - Support deployment of pre-developed PPM components to IÉ Power Platform environments (Dev/Test + Prod)
 - Data model configuration.
 - Workflow configuration using out-of-box logic.
 - Core security setup, role-based access, and Azure AD integration.
 - Deployment of standard dashboards, views, and reports.
 - Applying IÉ branding to the user experience.
 - Basic Microsoft 365 integrations as required.
 - Support with any data migration activities.
 - Pre Deployment Testing

- MVP Pilot Documentation

- Solution Design Document
- Security Model
- Data Dictionary
- Role-Based User Guides
- Deployment Guide
- Administrator Guide / Runbook / Operations Manual
- Configuration & Customisation Log
- Audit Logs
- Training Material Pack (for internal and system admin training)
- Support & Troubleshooting Guide

- Training

- End-user and pilot group training sessions
- System administrator training
- Delivery of materials, walkthroughs, and Q&A sessions

- MVP Pilot Support

After completion of the MVP Pilot CI will conduct an evaluation to determine its success. The purpose of this evaluation is to:

- Collate user feedback
- Assess system performance and usability
- Provide recommendations for:
 - proceeding with vanilla deployment,
 - enhancements and integration requirements,
 - change management and adoption needs.

Solution Enhancements and/or Full Deployment & Post Deployment Support (Optional)

The MVP Pilot evaluation will conclude if CI wish to proceed System Enhancements and/or Full Deployment & Post Deployment Support. The scope, scale, and specific deliverables within this phase will be determined by the findings, recommendations, and prioritised requirements identified as part of the MVP Pilot evaluation.

Thus, this phase consists of several independent optional components, each of which CI may trigger individually, in combination, or not at all.

CI will only activate the options that best address:

- The functional gaps identified during the pilot.
- Adoption challenges or feedback from end-users.
- Any technical limitations or enhancements needed to support full organisational use.
- Integration, reporting, or automation requirements not supported by the vanilla product.

Option A – Licencing and full Vanilla Deployment

- Licencing
 - Provision of licences to cover the full functionality of the vanilla ‘out of box’ solution for the full deployment as per the user numbers identified in section 2.2 Users.
 - The solution shall be deployed into an IÉ managed Microsoft Power Platform environment. All platform hosting, infrastructure, and Microsoft licensing requirements are the responsibility of IÉ and are expressly excluded from the scope of this tender. This includes, but is not limited to, the provision of a Dev/Test and Production Power Platform environments, Microsoft Power Apps Premium licensing and Power BI Premium/Pro licensing. The supplier is required to deliver and configure the solution within the IÉ environment only and will not be responsible for providing, managing, or procuring any underlying platform licences or hosting services.
- Support full deployment of all CI portfolio, programmes and projects to the PPM system.
- Provision of Full Deployment Technical Services
 - Implementation Approach & Plan
 - Provide IÉ Power Platform Requirements (Environment Specs, Licences etc.)
 - Support deployment of pre-developed PPM components to IÉ Power Platform environments (Dev/Test + Prod)
 - Data model configuration.
 - Workflow configuration using out-of-box logic.
 - Core security setup, role-based access, and Azure AD integration.
 - Deployment of standard dashboards, views, and reports.
 - Applying IÉ branding to the user experience.
 - Basic Microsoft 365 integrations as required.
 - Support with any data migration activities.

- Pre Deployment Testing
- Provision of the following documentation
 - Solution Design Document
 - Security Model
 - Data Dictionary
 - Role-Based User Guides
 - Deployment Guide
 - Administrator Guide / Runbook / Operations Manual
 - Configuration & Customisation Log
 - Audit Logs
 - Training Material Pack (for internal and system admin training)
 - Support & Troubleshooting Guide
- Provision of the following Training
 - End-user training sessions
 - System administrator training
 - Train the trainer type trainer to a select number of superusers
 - Delivery of materials, walkthroughs, and Q&A sessions

Option B - Configurations & Customisations to Vanilla ‘Out of box’ solution

- The provision of time-and-materials development resources to deliver enhancements to the vanilla ‘out of box’ solution. These enhancements may include but not limited to the following:
 - Additional workflows or automation
 - Data model extensions
 - Custom UI elements (forms, fields, business rules)
 - Custom dashboard/report development
 - Security model refinement
 - Project template changes
 - Additional user roles or permissions
 - Data migration services for wider rollout
- As part of this option the vender shall provide:
 - Delivery Plan for each enhancement.
 - Pre Deployment Testing.
 - Update ‘vanilla’ documentation as required.

Option C - Integration with corporate systems

- The provision of time-and-materials development resources and required licencing to deliver Integrations with IE corporate systems.
- Provide an integration between the PPM solution and IE's SAP ERP system. This includes providing a secure, scalable integration platform capable of supporting current SAP ECC

operations and future migration to SAP S/4HANA. The solution must handle high volumes of financial data, use modular and reusable integration components, and ensure accurate mapping between SAP cost objects and PPM project structures.

- Provide an integration between the PPM solution and IÉs scheduling tools (OPC, P6 and MS Project). This includes support for importing common scheduling file formats (such as P6, MS Project, Excel, XER, XML, XLSX and MPP) to create or update project schedules.
- Provide a Delivery Plan for each integration
- Provide documentation related to integration which may include but not limited to:
 - Integration Architecture Document
 - Solution Design Document
 - Data Mapping Specification
 - API / Interface Specifications
 - Security Design
 - SIT / UAT Test Packs
 - Deployment Guide
 - Support & Troubleshooting Guide
 - Configuration Log
 - Runbook / Operations Manual

Option D + Post Deployment Support

The provision of ongoing post-implementation assistance to ensure the long-term stability, performance, and adoption of the PPM solution. Under this phase, the supplier will deliver technical support, issue resolution, maintenance, minor configuration updates, release management, and general operational assistance to both end-users and system administrators. Support may also include onboarding assistance, periodic health checks, performance reviews, and user adoption guidance.

Support is required for both the standard out-of-the-box solution and any configurations, customisations, or integrations added as part of the solution rollout.

Appendix 1 – Requirements and Specifications

1. Solution Overview

The Tenderer should plan to provide a demonstration (limited to one hour) of their capabilities and the proposed solution for IÉ. This should:

- Demonstrate their general understanding of IÉ goals and objectives, the challenges it faces and the purpose of this Tender.
- Provide a brief description of the proposed solution, the overall architecture, and the technologies, products and services user.
- Confirm that the proposed solution can meet the requirements in this ITT (any significant gaps or issues must be described)
- Highlight the key strengths of the proposed solution.
- Describe the proposed solution's integration capabilities. Specificity reference should be made to the solutions capability to integrate with SAP ERP and common scheduling tools (P6 + Microsoft Projects) as per section 2.2 'Non-Functional Requirements'.
- Provide the future Product Road Map, product automation and artificial intelligence

2. Detailed Specification of Requirements

2.1 Functional Requirements

2.1.1 Centralised Project, Programme & Portfolio Management

The solution shall provide a centralised location for managing all projects, programmes, and portfolios.

ID	Requirement Description
FR-PPM-01	The solution shall allow for the creation and management of Projects, Programmes, and Portfolios.
FR-PPM-02	The solution shall allow for the creation and management of Project, Programme, and Portfolio metadata.
FR-PPM-03	The solution shall allow for the capture of the full lifecycle of pipeline projects and opportunities, enabling their assessment and prioritisation

2.1.2 Governance, Assurance and Compliance

The solution shall ensure all projects follow the organisation's rules, standards, and legal requirements. It should make oversight and quality checks easy, so decisions are clear and properly documented.

ID	Requirement Description
FR-Gov-01	The solution shall provide a structured workflow that guides users through the initiation, submission, review, and approval of new projects.
FR-Gov-02	The solution shall provide structured gate reviews that enable the governance, assessment, approval, and controlled progression of projects through defined lifecycle stages.
FR-Gov-03	The solution shall provide comprehensive audit logs that capture all key user actions, data changes, and system events to support governance, compliance, and traceability.

2.1.3 Schedule & Programme Management

The solution shall support schedule and programme management by planning, tracking, and coordinating project timelines, and by consolidating data from other planning tools in use.

ID	Requirement Description
FR-SP-01	The solution shall allow users to create and manage project schedules of differing complexity natively within the application.
FR-SP-02	The solution shall allow users to monitor project schedules.
FR-SP-03	The solution shall allow for effective schedule management at the programme and portfolio levels.

2.1.4 Financial & Funding Management

The solution shall manage project finances and funding by planning, tracking, and controlling costs.

ID	Requirement Description
FR-FF-01	The solution shall provide an out of box configurable financial calendar.
FR-FF-02	The solution shall allow for the capture, management, and modification of project funding sources across the project lifecycle.
FR-FF-03	The solution shall allow for the creation, management, and modification of Project Budgets across the project lifecycle.
FR-FF-04	The solution shall allow for the creation, management, and modification of Project Forecasts across the project lifecycle.
FR-FF-05	The solution shall allow for the creation, management, and modification of Project Cashflows. The solution shall plan, monitor, and control the timing and magnitude of cash inflows and outflows throughout the project lifecycle.
FR-FF-06	The solution shall allow for the monitoring of project financial metrics and/or KPIs across the project lifecycle.

2.1.5 Risk & Issue Management

The solution shall support risk and issue management by identifying, tracking, and coordinating responses to risks and issues.

ID	Requirement Description
FR-RI-01	The solution shall enable users to identify, document, categorise, and assign ownership for risks across Projects, Programmes, and Portfolios.
FR-RI-02	The solution shall provide structured tools for assessing likelihood, impact, scoring, and financial exposure of risks at all levels.
FR-RI-03	The solution shall support effective management of risks through responses, mitigations, actions, timeline tracking, and governance controls.
FR-RI-04	The solution shall provide dashboards, filtering, search, and reporting capabilities to support effective monitoring of risks across all levels.
FR-RI-05	The solution shall provide comprehensive issue identification, tracking, and resolution functionality at Project, Programme, and Portfolio levels, ensuring consistent logging, categorisation, ownership, and management.

2.1.6 Change Management

The solution shall support change management by capturing, assessing, and tracking change requests to ensure impacts are understood and decisions are properly controlled.

ID	Requirement Description
FR-CM-01	The solution shall support the controlled capture, categorisation, impact assessment, and documentation of change requests across Projects, Programmes, and Portfolios.

FR-CM-02	The solution shall support the tracking, decision recording, and oversight of change requests as they progress through configurable approval steps, ensuring transparent and controlled baseline updates
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2.1.7 Benefits Management

The solution shall support benefits management by enabling clear definition, tracking, and governance of benefits to ensure strategic value delivery.

ID	Requirement Description
FR-BM-01	The solution shall support the structured identification, planning, classification, and documentation of benefits at Project, Programme, and Portfolio levels.
FR-BM-02	The solution shall track the performance and progress of benefits over time, comparing planned versus actual realisation across all levels.

2.1.8 Timesheet Management

The solution shall support timesheet management by capturing, tracking, and reporting time spent on project & non-chargeable activities to enable accurate monitoring and cost control.

ID	Requirement Description
FR-TM-01	The solution shall enable users to accurately record chargeable and non-chargeable time against authorised activities using a configurable Irish calendar that incorporates official public holidays.
FR-TM-02	The solution shall enable authorised managers to review and approve recorded time against Projects, Programmes, Portfolios, or resource assignments before it is committed to tracking and reporting.

2.1.9 Resource Management

The solution shall support resource management by planning, allocating, and tracking resources to ensure the right people are assigned to the right work at the right time.

ID	Requirement Description
FR-RM-01	The solution shall provide comprehensive resource configuration and setup capabilities to support the full Capital Investments organisation, including internal staff, external suppliers, project teams, specialist resources, and non-labour assets. The system shall support consistent, scalable, and permission-controlled management of all resource classes.
FR-RM-02	The solution shall enable skill-based resource searching so users can define the skills required for Projects, Programmes, and Portfolios, and quickly identify suitable resources. The system shall allow searching the resource database by skill and availability to support efficient assignment.
FR-RM-03	The solution shall support assignment of resources to Projects. It shall enable flexible allocation, approval, notification, tracking, and history management to ensure structured and auditable resourcing.

2.1.10 Reporting & Analytics

The solution shall support reporting and analytics across portfolio, programme, and project levels by consolidating data and providing clear, timely insights to support decision-making.

ID	Requirement Description
FR-RA-01	The solution shall support periodic project status capture aligned to 13 four-week periods, with snapshot history and RAG/KPI monitoring to provide consistent visibility of project performance over time.
FR-RA-02	The solution shall deliver out-of-the-box dashboards that provide clear, real-time visibility across Projects, Programmes, and Portfolios, with configurable views that require no custom development. These dashboards shall support aggregated roll-up reporting and drill-down

	navigation, enabling users to move effortlessly from high-level strategic insight to detailed project data.
FR-RA-03	The solution shall include ready-made, configurable financial reports that require no custom development and can be tailored to different organisational, project, and funding needs. These reports shall support rapid, accurate financial insight across Projects, Programmes, and Portfolios.
FR-RA-04	The solution shall include ready-made, configurable schedule reports that require no custom development and support visual, portfolio-level insights into project timelines, milestones, critical paths, and performance. These reports shall be adaptable to organisation, programme, and stakeholder needs while leveraging real-time schedule data.
FR-RA-05	The solution shall provide out-of-the-box configurable reporting for Risks and Issues, enabling users at all levels to monitor exposure, status, trends, and escalation points across Projects, Programmes, and Portfolios. Reports shall be easily tailored without custom development and support decision-making, governance, and assurance.
FR-RA-06	The solution shall allow users to seamlessly extract structured data into external analytics tools such as Power BI so that Framework Clients can build their own bespoke reports, dashboards, and visualisations without relying on the vendor. This ensures flexibility, self-service reporting, and the ability to meet evolving organisational and government reporting needs.

2.1.11 Workflows

The solution shall support workflows across portfolio, programme, and project levels by automating approval steps, standardising processes, and ensuring consistent control and governance.

ID	Requirement Description
FR-WF-01	The solution shall provide fully configurable workflows that support governance, approvals, routing, and process automation across Projects, Programmes, Portfolios. These workflows shall be adaptable without custom development and support individual, group, role-based, and document-related approvals with full auditability.
FR-WF-01	The solution shall provide workflow governance capabilities that allow the PMO to monitor, track, and audit all approvals across Projects, Programmes, and Portfolios. This ensures compliance with organisational governance frameworks, approval rules, and audit requirements.

2.1.12 User Access & Security

The solution shall support user access and security across portfolio, programme, and project levels by controlling permissions, ensuring appropriate data access, and protecting information in line with organisational policies.

ID	Requirement Description
FR-UAS-01	The solution shall allow administrators to define and manage security roles that control what users can create, view, edit, or delete. Access must be governed through configurable role structures that support organisational governance and least-privilege principles.
FR-UAS-02	The solution shall provide workflow governance capabilities that allow the PMO to monitor, track, and audit all approvals across Projects, Programmes, and Portfolios. This ensures compliance with organisational governance frameworks, approval rules, and audit requirements.
FR-UAS-03	The solution shall integrate with Entra ID (Azure Active Directory) to enable single sign-on, identity governance, and strong authentication. This ensures secure login, centralised account control, and public-sector-compliant access management.
FR-UAS-04	The solution shall maintain comprehensive audit trails covering user activity, approvals, changes, and access events. This is essential for governance, compliance, assurance, and security monitoring.

2.2 Non-Functional Requirements

2.2.1 System Performance

The following requirements govern the performance expectations of the solution under expected operational conditions. Given the Power Platform delivery model, tenderers must address Dataverse throughput, Power Automate execution latency, and capacity planning in their response.

ID	Requirement Description
NFR-PERF-01	The solution shall load primary dashboards and portfolio views within 3 seconds under normal load conditions.
NFR-PERF-02	The solution shall support concurrent user access without degradation, supporting the full projected user base as defined in section 2.2.
NFR-PERF-03	Power Automate workflows shall complete non-interactive background processes within a defined SLA, with approval notifications triggered within 2 minutes of a qualifying event.
NFR-PERF-04	The solution shall handle Dataverse API call limits in line with Microsoft-published service protection limits without service interruption, implementing throttling and retry logic where required.

2.2.2 Data Management

The following requirements govern how project, programme, and portfolio data is stored, protected, managed, and accessed within the Power Platform environment. Data residency, sovereignty, and GDPR compliance are primary considerations for IÉ as a public-sector entity.

ID	Requirement Description
NFR-DM-01	All project, programme, and portfolio data shall be stored within Dataverse in IÉ's managed Power Platform environment, with no data residing on supplier-managed infrastructure.
NFR-DM-02	The solution shall support Dataverse data retention policies aligned with IÉ's records management obligations and GDPR requirements.
NFR-DM-03	The solution shall implement Dataverse column-level security where sensitive financial or personal data fields are present, in addition to row-level security.
NFR-DM-04	The supplier shall provide a complete data dictionary and entity relationship model for all Dataverse tables introduced or modified as part of the solution.
NFR-DM-05	The solution shall support data archiving and purging strategies, including configuration guidance, without impacting live system performance.
NFR-DM-06	All data extracts and API connections to external tools such as Power BI shall enforce role-based access controls consistent with the PPM security model.
NFR-DM-07	The solution shall not introduce custom connectors that bypass IÉ's Data Loss Prevention (DLP) policies as configured in the Power Platform Admin Centre.
NFR-DM-08	The supplier shall provide a data migration strategy, tooling, and validation plan to support the transition of any existing project data into the new solution.

2.2.3 Usability and Accessibility

The following requirements govern the usability and accessibility of the solution across the full user population, including office-based, site-based, and mobile users. Compliance with WCAG 2.1 Level AA is a mandatory expectation for IÉ as a public-sector body.

ID	Requirement Description
NFR-USE-01	The solution shall conform to WCAG 2.1 Level AA accessibility standards across all Power Apps interfaces.
NFR-USE-02	The solution shall be functional on Microsoft-supported browsers (Edge, Chrome) and shall support access via the Power Apps mobile application on iOS and Android.
NFR-USE-03	The solution shall provide contextual help, tooltips, and in-application guidance to support user adoption without requiring reference to external documentation.
NFR-USE-04	All data entry forms shall provide inline validation and meaningful error messages to guide users through correct data entry prior to submission.
NFR-USE-05	The user interface shall be configurable to reflect IÉ branding including colours, logos, and terminology, without requiring custom development.
NFR-USE-06	The solution shall provide a consistent navigation structure across Projects, Programmes, and Portfolios, minimising the number of interactions required to complete common tasks.

2.2.4 Power Platform

The following requirements are specific to the Microsoft Power Platform delivery model and address application lifecycle management (ALM), solution architecture, environment strategy, and platform governance. These requirements are critical to ensure the long-term supportability, upgradeability, and governance of the solution within IÉ's managed environment.

ID	Requirement Description
NFR-PP-01	The solution shall be delivered as a managed solution within Dataverse, enabling controlled deployment across Dev/Test and Production environments via standard ALM pipelines.
NFR-PP-02	The solution shall comply with Microsoft Power Platform best practices for solution layering, avoiding hard-coded environment URLs, connection references, and environment variables.
NFR-PP-03	All Power Automate flows shall use connection references and environment variables rather than hard-coded connections to support ALM and multi-environment deployment.
NFR-PP-04	The solution shall be compatible with the current generally available release of Power Platform, and the supplier shall define a process for managing Microsoft release wave updates (April and October cycles).
NFR-PP-05	The supplier shall document all Dataverse customisations, custom connectors, PCF controls, and Power Automate flows within the Configuration and Customisation Log, as a required deliverable.
NFR-PP-06	The solution shall not exceed IÉ's allocated Dataverse storage capacity, and the supplier shall provide tooling or guidance to monitor and manage storage consumption over the contract lifetime.
NFR-PP-07	The solution shall implement a Power Platform environment strategy aligned with Microsoft's recommended ALM framework, including use of Power Platform Pipelines or Azure DevOps for controlled deployment.
NFR-PP-08	All canvas Power Apps shall be built using the responsive layout engine and shall not rely on fixed pixel positioning.

2.2.5 SAP ERP Integration

IÉ operates SAP ECC for financial accounting and controlling. The PPM solution must receive actual cost data from SAP to support financial monitoring, budget variance, and cashflow management within the PPM platform.

ID	Requirement Description
NFR-INT-01	The integration shall support extraction of actual costs, commitments, and purchase orders from SAP FI/CO modules on both a scheduled and on-demand basis.

NFR-INT-02	The integration shall support mapping between SAP WBS elements, cost centres, and internal orders and the corresponding PPM project structures.
NFR-INT-03	The integration shall be implemented via a middleware or integration platform (e.g., Azure Integration Services, Azure Logic Apps, or SAP BTP Integration Suite) rather than direct point-to-point connection.
NFR-INT-04	The integration shall include error handling, retry logic, and alerting mechanisms for failed or incomplete data transfers.
NFR-INT-05	The supplier shall deliver an Integration Architecture Document, Data Mapping Specification, and API/Interface Specification as named deliverables.
NFR-INT-06	The integration architecture shall be designed to accommodate IE's planned migration from SAP ECC to SAP S/4HANA without requiring a full redesign of the integration layer.
NFR-INT-07	The system shall enable authorised users to upload finalised financial data for each 4-week reporting period into the solution using a standardised Excel template, prior to integration with SAP ERP is complete. The system shall perform data validation upon upload and ensure controlled access in accordance with financial governance requirements.

2.2.6 Scheduling Tool Integration (P6, OPC, MS Project)

Capital Investments currently utilises Oracle Primavera P6, OPC, and Microsoft Project as scheduling tools. The PPM solution must support integration with these platforms to consolidate schedule data within the portfolio management layer.

ID	Requirement Description
NFR-INT-08	The solution shall support import and export of schedules in XER (Primavera P6), MPP (MS Project), and XML formats as a minimum.
NFR-INT-09	On import, the solution shall validate schedule data for integrity, including missing predecessor links and constraint conflicts, and surface warnings to the user prior to committing the import.
NFR-INT-10	The integration shall preserve WBS hierarchy, milestone designations, and baseline data on import from external scheduling tools.
NFR-INT-11	The solution shall support scheduled or event-triggered synchronisation with external scheduling tools, in addition to manual file upload, where technically feasible.
NFR-INT-12	The system shall enable authorised users to upload finalised schedule data for each 4-week reporting period into the solution using a standardised Excel template, prior to integration with scheduling tools is complete. The system shall perform data validation upon upload and ensure controlled access in accordance with schedule governance requirements.

2.2.7 Microsoft 365 Integration

IE operates a Microsoft 365 environment including Teams, SharePoint Online, Outlook, and Entra ID (Azure Active Directory). The PPM solution shall leverage the existing M365 estate to support collaboration, document management, and identity governance.

ID	Requirement Description
NFR-INT-13	The solution shall integrate with Microsoft Teams to support approval notifications, project status updates, and workflow alerts via Teams channels or adaptive cards.
NFR-INT-14	The solution shall integrate with SharePoint Online for document management, enabling documents to be linked or stored against projects within a managed IE document library structure.
NFR-INT-15	The solution shall integrate with Outlook for approval notifications and, where applicable, calendar-based milestone visibility.

NFR-INT-16	The solution shall leverage Entra ID (Azure AD) group membership for dynamic security role assignment within the PPM solution where appropriate.
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2.2.8 Power BI Integration

The PPM solution shall support self-service reporting and analytics via Power BI, enabling IÉ's analytics team to build bespoke reports and dashboards from PPM data without vendor dependency. This aligns with the mandatory requirement FR-Man-05 (Data Extraction) as set out in the ITT.

ID	Requirement Description
NFR-INT-17	The solution shall expose Dataverse data via the Power BI Dataverse connector or a dedicated semantic layer to enable self-service reporting by IÉ's analytics team.
NFR-INT-18	The supplier shall provide a baseline Power BI dataset model aligned to the PPM data structure as a named deliverable within the MVP pilot.
NFR-INT-19	Embedded Power BI reports within the PPM solution shall respect the PPM role-based access control model using Power BI row-level security, consistent with the Dataverse security model.

2.2.9 Cyber Security

The tendered solution must conform with the Cybersecurity requirements. The tenderer must complete the associated document, IÉ Cybersecurity Requirements, and submit as part of the tender submission. Non-conformance will result in exclusion from this tender process.

3. Solution Delivery

The delivery of the solution the tenderer is proposing shall include services to ensure a smooth transition and adoption within IÉ CI.

The Tenderer shall describe their solution delivery approach and capabilities as below:

3.1 Project Plan, Deliverables, Dependencies and Governance Approach

The tenderer will be responsible for developing and testing all business functionality for the new solution as well as the solutions that support transitional processes, such as data migration and deployment of the solution into the IÉ's technology estate. This will include, and not limited to:

ID	Requirement Description	Included Y/N	Details
SDP-01	The tenderer shall provide a detailed project plan for their proposed implementation approach for both the MVP Pilot and Full deployment. These shall include timelines, budgets, milestones, assumptions, dependencies, roles and responsibilities, resource allocation, and a phased deployment schedule showing the planned date on which the system will be available.		
SDP-02	The tenderer shall provide details of their deliverables, acceptance criteria, phases of the roll out programme and review and approval processes.		
SDP-03	The tenderer shall describe their proposal for the implementation and system commissioning of the solution, including configuration and setup, provision of standard reports, training, data migration and		

	conversion, system testing, user acceptance testing and quality assurance.		
SDP-04	The tenderer shall describe their governance framework, including project oversight, decision making processes, compliance and reporting, stakeholder engagement, risk and risk mitigation and the monitoring of performance.		
SDP-05	The tenderer shall set out their approach to delivery of user training for IÉ staff during the implementation and installation phases to ensure the most cost-effective training program.		

3.2 Change Management Approach

The tenderer must demonstrate a structured and coherent approach to change management and solution maintenance during the implementation and over the lifetime of the agreement.

ID	Requirement Description	Included Y/N	Details
SDC-01	The tenderer must comply with IÉ change management policy, as available in Appendix 4 – Change Management Policy.		
SDC-02	The tenderer shall describe their method of change management, including information management and requirements, identification of change, change implementation, communication and collaboration, change monitoring and auditing, and feedback and continuous improvement. This change management should be as part of the tender's enablement of ISO 19650.		
SDC-03	The solution will have significant functional fit to the requirements using out of the box (or standard) functionality.		

3.3 Implementation Team and Service Manager

The implementation team and service manager are the team responsible for the successfully execution, management and support of the project and the relationship with IÉ over the longer term. The tenderer should demonstrate the capability, experience and structure that will drive the successful implementation and delivery of the solution.

The tenderer must include the following:

- An organisation chart which identifies the key resources with responsibility for the delivery of each of the key components of this Project.
- The participants of the implementation team, CVs for each resource and relevant Rail and Document Management solutions experience. Resources should include:
 - Implementation Project Manager
 - Technical Lead Architect
 - Service Manager
- An estimate of the resource requirements within IÉ, and their roles, to deliver the project.

The tenderer must ensure that these resources identified are available for delivery on this implementation project. All staff engaged by the tenderer in the project implementation, including subcontracted staff, must be vetted by IÉ.

Please Describe:

3.4 Technical Implementation

The technical implementation will be hosted within IÉs Power Platform.

- The Tenderer should provide a description of the technical implementation of their solution within IÉs Power Platform, including:
 - The process in which the Tenders solution is hosted within IÉs Power Platform environments.
 - Data Security (confirm how the data will be held in EU) and protection from malware
 - Backups and Disaster Recovery provision
 - Methodology for updates, testing and deployment
- A security questionnaire will need to be satisfactorily completed. The supplier should confirm they accept this condition of contract.

Please Describe:

4 Post Implementation, Support and Maintenance

IÉ requires a high-quality post implementation support and maintenance service, including service level agreements, knowledge transfer, exit strategy, change and release management and reporting system. It is expected that this support will be provided for both the standard ‘Out of Box’ functionality as well as any changes made to the vanilla product by the vendor.

The tenderer will provide help desk support relating to the specific configuration that has been installed within the IÉ Power platform environment. The tenderers will agree with IÉ the appropriate communications methods and customer support contacts.

The tenderer shall describe their post implementation, support and maintenance capabilities as below:

ID	Requirement Description	Included Y/N	Details
PS-01	It is expected the Tenderer will provide a warranty period for defect resolution, following the go live period. This period will be for a minimum agreed period, or until all defects that are identified during the go live period or materialise during the warranty period are resolved. Describe the warranty that will be provided.		
PS-02	The Tenderer must demonstrate a structured and coherent approach to upgrades and maintenance requirements over the lifetime of the agreement.		
PS-03	The Tender ensure the solution provided Power Platform solution shall be regularly reviewed and updated to ensure continued compatibility and alignment with IÉ's Power Platform version and standards.		
PS-04	Support service desk shall be available on telephone support, email support and web-based ticketing system and support the required service levels. The tender can provide details on their help desk service escalation and response levels.		
PS-05	The tenderer shall have a robust incident management process, including incident logging, categorisation, escalation and resolution. All incidents shall be tracked and documented in the tenderer's ticketing system, with regular status updates provided to IÉ.		
PS-06	The tenderer shall conduct regular reviews of tickets logged and identify problems that may exist, perform root cause analysis and proactively manage problems through to resolutions.		
PS-07	The tenderer shall comply with the documented service level objectives and will be reviewed every 12 months or as needed.		
PS-8	The Tenderer is required to perform penetration tests to identify any potential security vulnerabilities. The tenderer must correct any confirmed vulnerabilities. Provide details of containment and eradication of any threats.		
PS-9	The Tenderer must outline a robust Exit Management Plan for any planned/un-planned migration off the solution. This must include a detailed project plan and supporting methodology. An Exit Management plan is intended to support an orderly transition of the Services from the Contractor to IÉ and/or other Incoming Tenderer in the event of termination of all or part of this Agreement or its expiry.		

	It is required that the Tenderer detail an exit management plan which will, as a minimum: • Contain separate mechanisms for dealing with Ordinary Exit and Emergency Exit; • Set out the management structure to be employed during both transfer and cessation of the Services in an Ordinary Exit and an Emergency Exit; and • Contain a detailed description of both the transfer and cessation processes, including a timetable applicable in the case of an Ordinary Exit and an Emergency Exit. At the request of IÉ the Exit Plan shall anticipate a phased migration of the Services from the Tenderer to IÉ and/or other Incoming Supplier in such manner as IÉ shall require. • Document how the Services will transfer to IÉ and/or other Incoming Tenderer, including details of the processes, documentation, data transfer, systems migration, security and integration with legacy systems; • Address any other issues in the transition of the Services from the Tenderer to IÉ and/or other Incoming Tenderer's with the aim of ensuring that there is no disruption to or degradation of the Services during the Termination Assistance Period		
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5 Environmental, Social and Governance

1. IÉ is committed to sustainability and operating responsibly. The organisation is aware that ICT operations, including hosted applications, can potentially have a significant environmental and climatic impact.
2. The Tenderer should:
 - a. Provide a proposal outlining the specific measures to be adopted to address the environmental, social and economic impacts throughout the project delivery.
 - b. Highlight any innovations or unique solutions within their technical approach that set their proposal apart from others, explaining how these innovations will enhance the project's cost efficiency, performance or sustainability.
 - c. Indicate what measures it utilises to reduce its carbon footprint, for example the use of data centres utilising only renewable energy.

Responses in this section will be used to evaluate item 5 - Sustainability, CSR, and Environmental Responsibility as part of the award criteria.

Tenderer Response to Requirement

Please Describe:

Appendix 2 – Detailed Functional Requirements Specs

This section contains detailed specifications for each functional requirement. These specifications are intended as guidance to the vendor when responding to the functional requirements. Thus, these specifications are for information only. They should not be interpreted as additional or standalone requirements.

Centralised Project, Programme & Portfolio Management

Requirement	Spec
FR-PPM-01 Capture Projects, Programmes & Portfolios The solution shall allow for the creation and management of Projects, Programmes, and Portfolios.	- Allow for the creation of new Projects within an existing Programme and/or Portfolio. Or if required as a standalone project not associated with an existing Programme and/or Portfolio. - Allow for the creation of new Programmes as required. - Allow for the creation of new Portfolios as required. - The creation of Projects, Programmes or Portfolios are done using a standardised input form. Input forms should be configurable to have required and non-required fields. - Allow for the creation of relationships between Projects, Programmes and/or Portfolios.
FR-PPM-02 Capture Projects, Programmes and Portfolios Metadata The solution shall allow for the creation and management of Project, Programme, and Portfolio metadata.	- Provide standard and/or configurable metadata fields for Projects, Programmes, and Portfolios. - Support mandatory, optional, and conditional metadata fields. - Allow different metadata sets per entity type (e.g., Project vs Programme). - Allow metadata to be entered / edited based on user permissions and workflow stage. - Support metadata changes through configurable workflows (e.g., approval required for certain fields). - Provide field-level validation, lookup lists, and dropdowns. - Ensure metadata is accessible for reporting.
FR-PPM-03 Capture Project / Opportunity Identification The solution shall allow for the capture of the full lifecycle of pipeline projects and opportunities, enabling their assessment and prioritisation	- Standardised Capture of Opportunities via a digital input form. - Classify and score opportunities based on configurable criteria. - Assign opportunities standard and/or configurable metadata fields e.g. opportunity classification. - Support configurable review, assessment, and approval workflows. - Provide prioritisation tools. - Offer reporting dashboards showing pipeline volume, status, prioritisation etc.

Governance, Assurance and Compliance

Requirement	Spec
FR-Gov-01 Project Creation Workflow The solution shall provide a structured workflow that guides users through the initiation, submission, review, and approval of new projects.	• Enable creation of new project requests using configurable templates. • Allow for linking or attachment of supporting documents. • Provide automated routing of project requests through configurable predefined review and approval workflows. • Allow reviewers to evaluate, comment on, approve, pause or reject project requests with full audit trails. • Ensure mandatory data validation prior to project approval and creation in the system. • Convert approved requests into active Projects with all captured metadata transferred automatically.
FR-Gov-02 Gate Review Workflows The solution shall provide structured gate reviews that enable the governance, assessment, approval, and controlled progression of projects through defined lifecycle stages.	• Support configurable stage gate models tailored to different projects (e.g. Project Size, Project Type etc.) • Capture required documentation, metadata, and evidence needed for each gate. • Configurable workflows for gate submission, review, approval, pause, rejection, or rework. • Configurable scoring, commentary, and decision logging by reviewers and governance boards. • Maintain audit trails of gate decisions, approvers, dates, and justification. • Display gate status and progression within dashboards and project lifecycle views.
FR-Gov-03 Audit Logs The solution shall provide comprehensive audit logs that capture all key user actions, data changes, and system events to support governance, compliance, and traceability.	• Record changes to Projects, Programmes, Portfolios, financials, metadata, workflows, and permissions. • Capture user identity, timestamp, old value, new value, and the nature of the change. • Log key system events such as approvals, submissions, deletions, status changes, and role updates. • Support role-based access to audit logs, ensuring only authorised users (e.g., Admin/PMO) can view or manage them.

Schedule & Programme Management

Requirement	Spec
FR-SP-01 Create and Manage Project Schedules The solution shall allow users to create and manage project	• Provide a schedule configurable to a standardised Irish Calendar including Irish bank holiday periods or other non-working days. • Allow for the creation and modification of work breakdown structures.

schedules of differing complexity natively within the application.	• Define and modify project deliverables, milestones, activities, tasks and work packages • Capture, manage and modify start, finish dates (planned, forecasted & actual) for project deliverables, milestones, activities, tasks and work packages. • Capture, manage and modify floats for project deliverables, milestones, activities, tasks and work packages when required. • Capture, manage and modify internal dependencies - dependencies between two elements defined in the wbs structure (project deliverables, milestones, activities, tasks or work packages). • Capture, manage and modify project dependency constraints e.g. Finish-to-Start, Start-to-Start, Finish-to-Finish and Start-to-Finish • Capture, manage and modify project date constraints e.g. Start No earlier, Finish no later than, Must Start on and Must Finish on • Allow for the capture of multiple project schedule versions / baselines • Support multiyear schedules.
FR-SP-02 Project Schedule Monitoring The solution shall allow users to monitor project schedules.	• Provide scheduling visualisation in real-time graphical display (e.g. via Gantt charts, PERT charts) - which can be used to modify the project schedule. • Represent schedules in Days, Weeks, Months or Quarters • Allow for a captured baselined programme to be compared against the current forecast and actual project schedule. • Calculate and analyse project Critical Path • Compare dates, such as calculated dates and forecasted dates • Copy calculated dates to forecasted dates • Highlight when forecasts and actual completion dates approach planned targets • Allow a Rag status to be assigned to a milestone • Allow a status (complete, delayed etc.) to be assigned to project deliverables, milestones, activities, tasks and work packages • Allow comments / notes to be attached to project deliverables, milestones, activities, tasks and work packages • Export project schedule / scheduling visualisation to PDF • Allow for the export to external file types such as P6, MS Project and Excel. Common file formats such as XER, XML, XLSL and MPP etc. should be supported

FR-SP-03 Portfolio & Programme Schedule Management The solution shall allow for effective schedule management at the programme and portfolio levels.	• Allow for a rollup view of multiple projects schedules by portfolio or programme - providing a Capital investments master schedule. • Allow for the drill down from portfolio or programme view to associated child projects • Support centralised integrated scheduling across multiple projects and programmes - allowing for the creation and modification of dependencies and interdependencies between multiple projects. • Define and modify portfolio and programme deliverables, milestones, activities and tasks. • Provide a summarised view of child projects and their performance (Rag Status) of the related portfolio or programme.
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Financial & Funding Management

Requirement	Spec
FR-FF-01 Configurable Period Calendar The solution shall provide an out of box configurable financial calendar.	• A configurable period calendar that supports 4-week reporting periods. • Allow fiscal year start date, period definitions to be configurable. • Allow adjustments for leap years.
FR-FF-02 Funding Source Management The solution shall allow for the capture, management, and modification of project funding sources across the project lifecycle.	• Capture, manage and modify a single or multiple funding sources per project • Allow metadata related to funding sources to be captured, managed and modified. • Metadata fields should be configurable. Examples for metadata fields but not limited to are Funding Source Name, Funding type (e.g., internal, external, grant, loan, capital, operational), funding agency, validity period etc. • Categorise expenditure i.e. capital, externally funded, internally funded, additional revenue, existing revenue • Allow funding to be reallocated between projects when required - capturing original allocation, new allocation and justification for the change • Modifications of funding should allow for an approval process (workflow) if required • Ensure traceability and compliance by automatically logging changes thus creating an audit trail
FR-FF-03 Budget Management	• Capture project budgets based on estimates, funding, or historical data.

The solution shall allow for the creation, management, and modification of Project Budgets across the project lifecycle.	• Allow for alignment of project budget with project work breakdown structures (WBS), cost codes, and funding sources. • Allow for budget categories (e.g., labour, materials, overhead) • Allow for time-phased allocations (monthly, quarterly, 13-period) • Manage and modify project budgets due to scope changes, funding updates, or reforecasts etc. • Shall support multiple budget versions and baselines. Allow budgets to be modified due to scope changes, funding updates, or reforecasts etc. • Allow for updated budget version with change history / audit trail • Support multi-level budgets and hierarchical roll-ups • Allow for configurable budget approval workflows • Allow for hierarchical budgets
FR-FF-04 Forecast Management The solution shall allow for the creation, management, and modification of Project Forecasts across the project lifecycle.	• Capture project forecasts based on historical data, current progress, and planned activities. • Allow for configurable forecast modelling and parameters e.g. Forecast intervals (monthly, quarterly, 13-period), Forecast types (cost, revenue, resource) and Calculation methods (manual, linear projection, earned value) • Allow multiple forecasting models to be used per project • Allow for both top-down and bottom-up forecasting • Manage and modify project forecasts based on changes to the project e.g. change requests, updated assumptions, external factors etc. • Allow for updated forecast versions with change history / audit trail • Allow the accuracy of forecasting to be tracked over time • Allow for forecasting accuracy metrics, trend analysis, alerts for deviations • Allow multiple forecast scenarios to be modelled based on different assumptions • Provide a side-by-side comparison of forecast scenarios.
FR-FF-05 Cashflow Management The solution shall allow for the creation, management, and modification of Project Cashflows. The solution shall plan, monitor, and control the timing and magnitude of cash inflows and	• Capture planned project cash inflows e.g. expected cash receipts from funding sources, clients, or grants • Capture planned project cash outflows e.g. project activities or procurement • Allow the tracking of actual cash inflows and outflows from integrated ERP systems • Manage and modify project planned inflows and outflows

outflows throughout the project lifecycle.	• Provide future cashflows positions based on current trends and project progress • Allow for updated cashflow versions with change history / audit trail
FR-FF-06 Financial / Cost Monitoring The solution shall allow for the monitoring of project financial metrics and/or KPIs across the project lifecycle.	• Allow for configurable financial KPIs e.g. Cost Performance Index (CPI), Budget at Completion (BAC), Estimate at Completion (EAC), Return on Investment (ROI), Burn Rate etc. • Shall support scheduled and real-time data refresh • Allow financial metrics to be viewed in an interactive dashboard • Compare budgeted amounts with actual costs from ERP • Compare budgeted amounts with forecasted costs from ERP • Compare funding with budgeted amounts from ERP • Allow for variances between planned and actual cashflows to be captured • Provide Cashflow projections and liquidity curves • Share budget data with forecasting and cash flow modules. • Provide time-phased cash inflow projections • Provide time-phased cash outflow projections • Pull actuals from SAP Financial Accounting / Controlling ERP modules

Risk & Issue Management

Requirement	Spec
FR-RI-01 Risk Capture The solution shall enable users to identify, document, categorise, and assign ownership for risks across Projects, Programmes, and Portfolios.	• Allow capture of risks at Project, Programme, and Portfolio levels, including threats and opportunities. • Enable risk entry at any phase of the lifecycle, from initiation through closure. • Automatically generate a unique identifier for every risk using a standard naming convention. • Support configurable risk categories aligned to organisational taxonomy. • Allow assignment of risk owners and contributors at all entity levels. • Allow risks to be linked to project workstreams, WBS elements, or rolled up to parent entities (e.g., programme → portfolio). • Provide configurable templates for risk capture based on project types, categories, or bands. • Allow users to attach comments, notes, supporting documents, and evidence.

FR-RI-02 Risk Assessment The solution shall provide structured tools for assessing likelihood, impact, scoring, and financial exposure of risks at all levels.	• Support a configurable probability × impact risk matrix suitable for Projects, Programmes, and Portfolios. • Allow capture of current risk score and target risk score. • Capture estimated financial cost/exposure where applicable. • Support standardised risk assessment templates based on project or programme type. • Allow users to document assessment decisions, including acceptance, escalation, or deferral. • Maintain consistency in assessment scoring across all levels of the organisation.
FR-RI-03 Risk Management The solution shall provide comprehensive issue identification, tracking, and resolution functionality at Project, Programme, and Portfolio levels, ensuring consistent logging, categorisation, ownership, and management.	• Support capture of risk responses (e.g., Reduce, Avoid, Transfer, Accept). • Enable users to record mitigation actions, contingency plans, and progress updates. • Maintain an automated audit history of all changes (scores, mitigation updates, owner changes, dates). • Provide pre-mitigation and post-mitigation risk scoring for comparison. • Allow risks to be time-bound with tracking for creation date, review cycles, closure dates, and follow-ups. • Support alerts or notifications for overdue reviews, threshold breaches, or escalation conditions (with user ability to disable). • Allow risk owners to escalate risks upward or delegate downward across project, programme, and portfolio levels. • Enable linking or delegation of risks between programme/portfolio and their associated projects. • Allow addition of comments, notes, and file attachments as part of ongoing management.
FR-RI-03 Risk Monitoring The solution shall provide dashboards, filtering, search, and reporting capabilities to support effective monitoring of risks across all levels.	• Provide dashboards, lists, or matrices for viewing live risks at Project, Programme, and Portfolio levels. • Allow filtering by attributes such as top risks, high-scoring risks, overdue risks, category, or owner. • Identify overdue risks or those breaching probability/impact thresholds. • Support configurable RAG status rules to visually flag high-priority or overdue risks. • Allow exporting of risks into Excel, PDF, or other formats for governance reporting. • Enable visibility and filtering of child-entity risks at portfolio and programme levels (e.g., view programme + project risks combined).

	Support high-level and drill-down navigation between risk layers (Portfolio > Programme > Project).
FR-RI-03 Issues Management The solution shall provide comprehensive issue identification, tracking, and resolution functionality at Project, Programme, and Portfolio levels, ensuring consistent logging, categorisation, ownership, and management.	- Provide an Issues Log at Project, Programme, and Portfolio levels to record, track, and monitor issues throughout their lifecycle. - Assign each issue a unique system-generated identifier to ensure traceability and auditability. - Support configurable issue categories (e.g., operational, financial, vendor, technical) to enable structured classification. - Capture detailed issue information including description, impact, owners, due dates, resolution plans, and final decisions. - Provide configurable issue templates aligned to project types or categories to standardise issue capture. - Maintain issue history and status updates to support ongoing monitoring, reporting, and governance reviews.

Change Management

Requirement	Spec
FR-CM-01 Change Capture and Assessment The solution shall support the controlled capture, categorisation, impact assessment, and documentation of change requests across Projects, Programmes, and Portfolios.	- Enable users to identify and document change requests affecting baseline elements such as scope, schedule, finance, quality, and benefits. - Assign each change request a unique system-generated identifier using a standard naming convention. - Provide configurable change categories to classify and organise different types of changes. - Allow users to assess and record the impact of proposed changes on KPIs, including time, cost, quality, and business case benefits. - Allow users to capture supporting context, rationale, attachments, and commentary related to the change request.
FR-CM-02 Change Management Approval Workflow The solution shall support the tracking, decision recording, and oversight of change requests as they progress through configurable	- Track change requests as they move through defined, configurable approval stages, including visibility of the current status and next steps. - Maintain a complete record of approval decisions, including decision status (e.g., approved, rejected, cancelled) and any associated comments. - Enable configuration of decision statuses to align with organisational governance and change control processes.

approval steps, ensuring transparent and controlled baseline updates	• Provide alerts or notifications when approval actions, reviews, or decisions are overdue or pending. • Produce a detailed change approval report showing what changes were made, who made them, who approved them, and when decisions occurred. • Associate approved changes with the relevant baseline version and update baseline records accordingly once approval is completed.
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Benefits Management

Requirement	Spec
FR-BM-01 Benefits Identification, Planning & Classification The solution shall support the structured identification, planning, classification, and documentation of benefits at Project, Programme, and Portfolio levels.	• Enable users to identify, document, and define benefits including purpose, description, rationale, and expected value. • Support classification of benefits (e.g., financial, operational, customer, compliance, strategic) to standardise reporting. • Allow alignment of benefits to organisational strategic objectives, outcomes, and business case goals. • Capture planned benefit metrics, baselines, and expected timelines for realisation. • Enable assignment of benefit owners responsible for planning and delivery of each benefit. • Allow supporting commentary, assumptions, dependencies, and attachments to be recorded for each benefit.
FR-BM-02 Benefits Realisation Tracking The solution shall track the performance and progress of benefits over time, comparing planned versus actual realisation across all levels.	• Allow periodic capture of actual benefit values against planned baselines and forecasts. • Provide visibility of realised, forecasted, outstanding, and at-risk benefits through dashboards and reporting. • Highlight deviations from planned targets, including overdue, delayed, or under-performing benefits using RAG or threshold indicators. • Maintain a full historical record of updates, including changes to benefit forecasts, assumptions, and metrics. • Support aggregation and roll-up of benefit data from Projects to Programmes and Portfolios for consolidated reporting. • Allow export of benefit data for governance reviews, business case validation, and performance reporting.

Timesheet Management

Requirement	Spec
FR-TM-01 Time Recording Input The solution shall enable users to accurately record chargeable and non-chargeable time against authorised activities using a configurable Irish calendar that incorporates official public holidays.	• Allow users to record time against chargeable activities such as Projects, Programmes, Portfolios, and assigned overheads. • Support recording of non-chargeable time including annual leave, bank holidays, sickness, and other absence categories. • Use a configurable Irish calendar, including all official Irish public/bank holidays, to ensure correct non-working day allocation. • Ensure users can only enter time against activities they are assigned to or authorised for. • Provide configurable time entry periods (e.g., weekly, monthly) aligned with organisational policy. • Support validation rules including maximum daily hours, total timesheet balancing, and mandatory field checks. • Allow comments or notes to be added to individual time entries where clarification is needed. • Support coding or classification of time (e.g., billable/non-billable, internal/external) for reporting and finance alignment. • Provide visibility of planned absences (e.g., approved annual leave) to prevent conflicting time entries. • Allow timesheet data to be exported or accessed via reporting for financial reconciliation, utilisation analysis, and governance reviews.
FR-TM-02 Time Recording Approval The solution shall enable authorised managers to review and approve recorded time against Projects, Programmes, Portfolios, or resource assignments before it is committed to tracking and reporting.	• Allow Project, Programme, Portfolio Managers or Resource Managers (as defined by the client's workflow rules) to review time recorded against their assigned activities. • Provide visibility of all submitted time entries for validation before approval, including chargeable and non-chargeable categories. • Enable approvers to validate accuracy, completeness, and appropriateness of recorded hours prior to approval. • Allow approvers to request clarification or corrections from users before approving time entries. • Flag discrepancies, missing entries, or time recorded against unauthorised activities during the approval process. • Once approved, update time entries into utilisation, financial, and progress tracking reports.

Resource Management

Requirement	Spec
FR-RM-01 Resource Configuration The solution shall provide comprehensive resource configuration and setup capabilities to support the full Capital Investments organisation, including internal staff, external suppliers, project teams, specialist resources, and non-labour assets. The system Shall support consistent, scalable, and permission-controlled management of all resource classes.	• The solution must allow configuration of all internal staff as resources, including roles, availability, cost rates, and organisational attributes. • It shall support setup of external individuals, teams, and organisations as resources, with full capability for assignment, cost modelling, and contract alignment. • It shall enable configuration of non-labour resources, including equipment, vehicles, IT assets, and other physical resources, with calendars, rates, and utilisation rules. • The solution shall support role-based resource planning, allowing planning by generic role before assigning named individuals. • It shall provide skill-based resource configuration, enabling skills catalogues, proficiency levels, and matching of skills to project requirements.
FR-RM-02 Resource Search The solution shall enable skill-based resource searching so users can define the skills required for Projects, Programmes, and Portfolios, and quickly identify suitable resources. The system shall allow searching the resource database by skill and availability to support efficient assignment.	• Users shall be able to define the skills needed for Projects, Programmes, and Portfolios and use these requirements to drive resource planning. • The solution shall enable searching the full resource database by skill, including proficiency and qualifications. • The system shall allow users to identify resources who match required skills and are available, supporting effective scheduling and allocation.
FR-RM-03 Resource Management The solution shall support assignment of resources to Projects. It shall enable flexible allocation, approval, notification, tracking, and history management to ensure structured and auditable resourcing.	• Enable resource assignment • Allow resources to be allocated for defined time periods • Support reassignment of resources while retaining full historical audit trails, including previous assignees and durations. • Generate notifications to assigned team members and resource managers when new assignments occur or change. • Provide configurable multi-level approval workflows for resource assignments, • Track actual hours spent on Projects, Programmes, Portfolios, and individual tasks, while also tracking allocated time and linking both to work breakdown structures. • Allow resources to view their own allocations, and allow resource managers to view commitments, capacity, utilisation, and over-allocation risks for all resources under their responsibility.

Reporting & Analytics

FR-RA-01 Project Status Capture The solution shall support periodic project status capture aligned to 13 four-week periods, with snapshot history and RAG/KPI monitoring to provide consistent visibility of project performance over time.	• Allow for the capture of a periodic update / snapshot at each entity level (Projects, Programmes, and Portfolios) every four-week reporting period. • Provide configurable non-editable input fields that extract system-generated, for the period. • Provide configurable user input fields for narrative commentary related to the system data, including explanations of variances or highlights. • Provide configurable user input fields for summary commentary text boxes for overall period-level updates at each entity level. • Allow configurable validation rules to user input fields (e.g., maximum character limits, required fields). • Maintain a historical record of all period snapshots for comparison, governance, and reporting.
FR-RA-02 Out of Box Dashboard Reporting The solution shall deliver out-of-the-box dashboards that provide clear, real-time visibility across Projects, Programmes, and Portfolios, with configurable views that require no custom development. These dashboards shall support aggregated roll-up reporting and drill-down navigation, enabling users to move effortlessly from high-level strategic insight to detailed project data.	• Provide pre-configured dashboards for Projects, Programmes, and Portfolios, displaying key performance indicators (KPIs) such as schedule status, financial performance, risks, issues, milestones, and benefits. • Support roll-up aggregation, enabling Programme dashboards to automatically summarise underlying Project data, and Portfolio dashboards to summarise all included Programmes and Projects. • Enable drill-down navigation from Portfolio > Programme > Project > Task/Work Package levels, ensuring users can move from overview to detail in a single interface. • Shall support filtering by project, program, portfolio • Offer role-based dashboard views, ensuring Executives, Programme Managers, PMs, Planners, Finance, Commercial, and PMO teams see information that matches their responsibilities and access rights.
FR-RA-03 Portfolio & Programme Financial / Cost Management Reporting The solution shall include ready-made, configurable financial reports that require no custom development and can be tailored to different organisational, project, and funding needs. These reports shall support rapid, accurate	• Provide an out of box report that provides key project financial metrics • Support multi-level reporting across Projects, Programmes, Portfolios, including roll-up and drill-down capability. • Enable export to PDF or Excel for further analysis or external submission. • Provide role-based views so Project Managers, Finance, Programme Managers, and Executives see financial information appropriate to their responsibilities. • Shall support filtering by project, program, portfolio

financial insight across Projects, Programmes, and Portfolios.	
FR-RA-04 PPM Schedule Reporting The solution shall include ready-made, configurable schedule reports that require no custom development and support visual, portfolio-level insights into project timelines, milestones, critical paths, and performance. These reports shall be adaptable to organisation, programme, and stakeholder needs while leveraging real-time schedule data.	• Provide pre-configured schedule reports covering key metrics such as planned vs actual dates, milestone status, critical path indicators, baseline variance, and schedule health. • Support multi-level reporting across Projects, Programmes, Portfolios, including roll-up and drill-down capability. • Enable export to PDF or Excel for further analysis or external submission. • Offer role-based reporting views, giving Planners, Project Managers, Programme Managers, PMO, and Executives visibility appropriate to their oversight level.
FR-RA-05 Risk & Issues Reporting The solution shall provide out-of-the-box configurable reporting for Risks and Issues, enabling users at all levels to monitor exposure, status, trends, and escalation points across Projects, Programmes, and Portfolios. Reports shall be easily tailored without custom development and support decision-making, governance, and assurance.	• Provide pre-configured risk and issue reports, including top risks, overdue items, RAG indicators, exposure scoring, mitigation progress, and trend analysis across Projects, Programmes, and Portfolios. • Support roll-up and drill-down reporting, enabling consolidated Portfolio/Programme-level views as well as detailed Project-level insights. • Enable export to PDF or Excel for further analysis or external submission. • Offer role-specific reporting views for PMs, Programme Managers, PMO, Safety, Finance, and Executives, ensuring each stakeholder sees relevant risk and issue information
FR-RA-06 Data Extraction The solution shall allow users to seamlessly extract structured data into external analytics tools—such as Power BI allowing users to build their own bespoke reports, dashboards, and visualisations without relying on the vendor. This ensures flexibility, self-service reporting, and the ability to meet evolving organisational and government reporting needs.	• Provide standard connectors or APIs enabling secure extraction of solution data into external reporting tools such as Power BI, without requiring custom development. • Provide role-based controls ensuring only authorised users can access or export data.

Workflows

Requirement	Spec
FR-WF-01 Configurable Workflows The solution shall provide fully configurable workflows that support governance, approvals, routing, and process automation across Projects, Programmes, Portfolios. These workflows shall be adaptable without custom development and support individual, group, role-based, and document-related approvals with full auditability.	• Allow assignment of workflow steps for individual approvals, enabling precise routing to named resources. • Support multi-user or group approval steps, including sequential, parallel, or unanimous approval models. • Enable role-based workflow steps, where approvals are routed automatically to users performing a defined role (e.g., PM, Programme Manager, Commercial Lead). • Provide workflow configuration for process steps with target completion dates, supporting end-to-end process mapping and overdue step escalation. • Support configurable workflows for both process approvals and document approvals, including version control and audit trails. • Allow delegation of approval authority to another resource or role, with tracking of who delegated, to whom, and why. • Enable custom configuration of routing logic, conditions, branching, notifications, and escalation rules without custom development. • Allow Framework Clients to configure multi-level approval chains, including cross-Client approval if resources are shared across pan-government programmes. • Provide real-time workflow visibility, enabling users to see pending steps, approvers, status, and expected completion dates. • Generate automated notifications and reminders to approvers, contributors, and managers when workflow steps are triggered, assigned, overdue, or completed.
FR-WF-02 Workflow Governance The solution shall provide workflow governance capabilities that allow the PMO to monitor, track, and audit all approvals across Projects, Programmes, and Portfolios. This ensures compliance with organisational governance frameworks, approval rules, and audit requirements.	• Provide PMO-level visibility of all workflow steps, approvals, pending actions, bottlenecks, and overdue items across the entire Portfolio to ensure process compliance. • Enable audit-ready tracking and reporting of who approved what, when, and in what sequence, with full version history and justification where required. • Support governance enforcement, allowing the PMO to monitor adherence to mandatory workflows (e.g., gate reviews, CR approvals, document approvals) and flag deviations or non-compliance.

Appendix 3 – Non-Functional Requirements Rational

Non - Functional Requirements			
Category	ID	Description	Notes / Rationale
System Performance	NFR-Pref-01	The solution shall load primary dashboards and portfolio views within 3 seconds under normal load conditions.	Applies to both model-driven and canvas Power Apps.
	NFR-Pref-02	The solution shall support concurrent user access without degradation, supporting the full projected user base as defined in section 2.2.	Dataverse API limits apply per environment; supplier to confirm capacity approach.
	NFR-Pref-03	Power Automate workflows shall complete non-interactive background processes within a defined SLA, with approval notifications triggered within 2 minutes of a qualifying event.	Cloud flow execution latency must be specified and tested during pilot.
	NFR-Pref-04	The solution shall handle Dataverse API call limits in line with Microsoft-published service protection limits without service interruption, implementing throttling and retry logic where required.	Relevant where bulk operations, imports, or integrations are triggered.
	NFR-Pref-05	Power BI reports embedded within the solution shall refresh on a schedule no less frequent than every 4 hours during business hours.	Relevant for financial and portfolio dashboard reporting.
	NFR-Pref-06	The supplier shall define and document expected performance baselines for peak usage periods, including period-end reporting cycles aligned to the 13 four-week reporting calendar.	Period-end load is a foreseeable peak usage scenario.
Data Management	NFR-DM-01	All project, programme, and portfolio data shall be stored within Dataverse in IE's managed Power Platform environment, with no data residing on supplier-managed infrastructure.	Data sovereignty and residency requirement.
	NFR-DM-02	The solution shall support Dataverse data retention policies aligned with IE's records management obligations and GDPR requirements.	Retention policies configurable per entity type.
	NFR-DM-03	The solution shall implement Dataverse column-level security where sensitive financial or personal data fields are present, in addition to row-level security.	Extends beyond standard row-level security controls.
	NFR-DM-04	The supplier shall provide a complete data dictionary and entity relationship model for all Dataverse tables introduced or modified as part of the solution.	Required deliverable for long-term supportability and exit strategy.

	NFR-DM-05	The solution shall support data archiving and purging strategies, including configuration guidance, without impacting live system performance.	Critical given multi-year project portfolio data volumes.
	NFR-DM-06	All data extracts and API connections to external tools such as Power BI shall enforce role-based access controls consistent with the PPM security model.	Prevents data leakage via the reporting and analytics layer.
	NFR-DM-07	The solution shall not introduce custom connectors that bypass IÉ's Data Loss Prevention (DLP) policies as configured in the Power Platform Admin Centre.	Supplier must review and comply with existing DLP policy.
	NFR-DM-08	The supplier shall provide a data migration strategy, tooling, and validation plan to support the transition of any existing project data into the new solution.	Includes reconciliation and sign-off procedures.
Usability and Accessibility	NFR-USE-01	The solution shall conform to WCAG 2.1 Level AA accessibility standards across all Power Apps interfaces.	Legal and public-sector obligation; applies to canvas and model-driven apps.
	NFR-USE-02	The solution shall be functional on Microsoft-supported browsers (Edge, Chrome) and shall support access via the Power Apps mobile application on iOS and Android.	Supports site-based and field-based users.
	NFR-USE-03	The solution shall provide contextual help, tooltips, and in-application guidance to support user adoption without requiring reference to external documentation.	Reduces reliance on formal training for routine tasks.
	NFR-USE-04	All data entry forms shall provide inline validation and meaningful error messages to guide users through correct data entry prior to submission.	Reduces data quality issues and support overhead.
	NFR-USE-05	The user interface shall be configurable to reflect IÉ branding including colours, logos, and terminology, without requiring custom development.	Out-of-box theming capability.
	NFR-USE-06	The solution shall provide a consistent navigation structure across Projects, Programmes, and Portfolios, minimising the number of interactions required to complete common tasks.	UX efficiency standard relevant to daily users.
Power Platform	NFR-PP-01	The solution shall be delivered as a managed solution within Dataverse, enabling controlled deployment across Dev/Test and Production environments via standard ALM pipelines.	Prevents unmanaged component sprawl and supports governance.

NFR-PP-02	The solution shall comply with Microsoft Power Platform best practices for solution layering, avoiding hard-coded environment URLs, connection references, and environment variables.	Ensures portability across environments and upgrade compatibility.
NFR-PP-03	All Power Automate flows shall use connection references and environment variables rather than hard-coded connections to support ALM and multi-environment deployment.	Standard Microsoft ALM requirement.
NFR-PP-04	The solution shall be compatible with the current generally available release of Power Platform, and the supplier shall define a process for managing Microsoft release wave updates (April and October cycles).	Proactive management of platform updates.
NFR-PP-05	The supplier shall document all Dataverse customisations, custom connectors, PCF controls, and Power Automate flows within the Configuration and Customisation Log, as a required deliverable.	Supports supportability, handover, and exit strategy.
NFR-PP-06	The solution shall not exceed IÉ's allocated Dataverse storage capacity, and the supplier shall provide tooling or guidance to monitor and manage storage consumption over the contract lifetime.	Dataverse storage is capacity-limited by licence tier.
NFR-PP-07	The solution shall implement a Power Platform environment strategy aligned with Microsoft's recommended ALM framework, including use of Power Platform Pipelines or Azure DevOps for controlled deployment.	Deployment governance and change control.
NFR-PP-08	All canvas Power Apps shall be built using the responsive layout engine and shall not rely on fixed pixel positioning.	Ensures cross-device and cross-resolution compatibility.

Integration Requirements			
Category	ID	Description	Notes / Rationale
SAP ERP Integration	NFR-INT-01	The integration shall support extraction of actual costs, commitments, and purchase orders from SAP FI/CO modules on both a scheduled and on-demand basis.	Core financial actuals feed into PPM cost monitoring.

	NFR-INT-02	The integration shall support mapping between SAP WBS elements, cost centres, and internal orders and the corresponding PPM project structures.	Alignment of cost objects between systems.
	NFR-INT-03	The integration shall be implemented via a middleware or integration platform (e.g., Azure Integration Services, Azure Logic Apps, or SAP BTP Integration Suite) rather than direct point-to-point connection.	Supports future SAP ECC to S/4HANA migration without full redesign.
	NFR-INT-04	The integration shall include error handling, retry logic, and alerting mechanisms for failed or incomplete data transfers.	Operational resilience requirement.
	NFR-INT-05	The supplier shall deliver an Integration Architecture Document, Data Mapping Specification, and API/Interface Specification as named deliverables.	Required documentation for governance and supportability.
	NFR-INT-06	The integration architecture shall be designed to accommodate IÉ's planned migration from SAP ECC to SAP S/4HANA without requiring a full redesign of the integration layer.	Future-proofing requirement.
	NFR-INT-07	The system shall enable authorised users to upload finalised financial data for each 4-week reporting period into the solution using a standardised Excel template, prior to integration with SAP ERP is complete. The system shall perform data validation upon upload and ensure controlled access in accordance with financial governance requirements.	Allow financial data to be mass uploaded at the pilot stage.
Scheduling Tool Integration (P6, OPC, MS Project)	NFR-INT-08	The solution shall support import and export of schedules in XER (Primavera P6), MPP (MS Project), and XML formats as a minimum.	Aligns with scheduling tools currently in use by CI.
	NFR-INT-09	On import, the solution shall validate schedule data for integrity, including missing predecessor links and constraint conflicts, and surface warnings to the user prior to committing the import.	Data quality control on schedule import.
	NFR-INT-10	The integration shall preserve WBS hierarchy, milestone designations, and baseline data on import from external scheduling tools.	Ensures schedule fidelity across systems.
	NFR-INT-11	The solution shall support scheduled or event-triggered synchronisation with external scheduling tools, in addition to manual file upload, where technically feasible.	Reduces manual effort for schedule maintenance.
	NFR-INT-12	The system shall enable authorised users to upload finalised schedule data for each 4-week reporting period into the solution using a standardised Excel template, prior to integration with scheduling tools is complete. The system shall perform data validation upon upload and ensure controlled access in	Allow schedule data to be mass uploaded at the pilot stage.

		accordance with schedule governance requirements.	
Microsoft 365 Integration	NFR-INT-13	The solution shall integrate with Microsoft Teams to support approval notifications, project status updates, and workflow alerts via Teams channels or adaptive cards.	Meets users where they already work.
	NFR-INT-14	The solution shall integrate with SharePoint Online for document management, enabling documents to be linked or stored against projects within a managed IÉ document library structure.	Leverages existing M365 investment.
	NFR-INT-15	The solution shall integrate with Outlook for approval notifications and, where applicable, calendar-based milestone visibility.	Supports approvers who manage actions via email.
	NFR-INT-16	The solution shall leverage Entra ID (Azure AD) group membership for dynamic security role assignment within the PPM solution where appropriate.	Reduces manual role management overhead.
Power BI Integration	NFR-INT-17	The solution shall expose Dataverse data via the Power BI Dataverse connector or a dedicated semantic layer to enable self-service reporting by IÉ's analytics team.	Aligns with FR-Man-05 Data Extraction requirement.
	NFR-INT-18	The supplier shall provide a baseline Power BI dataset model aligned to the PPM data structure as a named deliverable within the MVP pilot.	Enables IÉ to begin self-service reporting from day one.
	NFR-INT-19	Embedded Power BI reports within the PPM solution shall respect the PPM role-based access control model using Power BI row-level security, consistent with the Dataverse security model.	Prevents data leakage via embedded reporting.

Appendix 4 – Change Management Policy



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Management PolicyV